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Learnable Frequency Domain Phase Residual and Graph-Contrastive Dynamic Convolution Network for Hyperspectral Anomaly Detection

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***Abstract*—Hyperspectral anomaly detection aims to identify sparse pixels with significant spectral or spatial differences from the surrounding background without prior target spectral information, and it has significant application value in key fields such as military reconnaissance and environmental monitoring. Existing methods suffer from insufficient frequency-domain feature extraction, static multiscale spatial modeling, and inadequate spectral–spatial feature interaction mechanisms. These limitations hinder their ability to adapt to complex scenarios involving target spectral variations, diverse target scales, and intricate spatial structures. Therefore, we propose a learnable frequency domain phase residual and graph-contrastive dynamic convolution (LFGDC) network for hyperspectral anomaly detection that integrates learnable frequency-domain modeling, multiscale dynamic convolution, and graph-structured contrastive constraints. First, a learnable frequency domain phase residual block (LFPRB) is introduced to explicitly exploit spectral frequency-domain characteristics. Second, a multiscale dynamic convolution module (MDCM) generates input-dependent convolution kernels to effectively adapt to spatial representations of targets at different scales. Finally, a graph-contrastive structure fusion module (GCSFM) is constructed to uncover global-local topological correlations and enhance feature discriminability. This is combined with adaptive frequency-domain modeling and multiscale feature fusion strategies to bolster model robustness. Extensive experimental results demonstrate that this method achieves superior detection performance compared to baseline methods on both public and produced datasets, providing an effective solution for hyperspectral anomaly detection in complex scenarios.**

***Index Terms*—Hyperspectral anomaly detection, learnable frequency domain phase residual block, multiscale dynamic convolution module, graph-contrastive learning.**

I. Introduction

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HYPERSPECTRAL remote sensing technology captures the spectral reflection characteristics of ground objects in the continuous narrow band range of 400-2500 nm, which can finely depict the differences in material composition of ground objects and provide unprecedented data support for ground object classification, target recognition, and anomaly detection. Hyperspectral anomaly detection (HAD), as one of the core tasks in this field, aims to automatically identify sparse pixels with significant spectral or spatial differences from the surrounding background without prior target spectral information [1]. With the unique advantage of unsupervised detection, this technology has irreplaceable application value in key fields such as military reconnaissance [2], [3], environmental monitoring [4], [5], resource exploration [6], and disaster emergency response [7]. The development of HAD technology has primarily focused on two core aspects: enhancing feature modeling capability and improving scene adaptability, leading to three main categories of approaches: statistics-based methods, representation-based methods, and deep learning-based methods.

Statistics-based methods are the most classic category in HAD. Their core idea is to establish a statistical model for the background and identify pixels that do not conform to this statistical distribution as anomalies. Global statistical methods are based on the Reed-Xiaoli (RX) algorithm [8], which assumes that the background follows a multivariate Gaussian distribution and realizes anomaly discrimination by calculating the Mahalanobis distance between pixels and the background mean. This method is only suitable for scenes with uniform backgrounds. Local statistical methods introduce a sliding window mechanism to cope with background inhomogeneity. Local RX [9] uses an inner-outer double window structure, where the inner window is used to estimate local background statistical characteristics and the outer window is used to exclude potential anomaly interference. Statistics-based methods rely on manually designed statistics and strong distribution assumptions, lacking robustness to complex factors such as spectral variation and band redundancy. In the complex background of actual hyperspectral images (HSI), their detection performance is easily severely affected, making it difficult to meet the requirements of high-precision detection.

Representation-based methods assume that the background can be linearly or nonlinearly represented by a set of basis vectors or dictionary atoms, while anomalies cannot be effectively represented, thereby identifying anomalies through representation residuals. Sparse representation methods [10] emphasize that background pixels can be sparsely represented under an overcomplete dictionary. Collaborative representation methods [11], [12] assume that background pixels can be linearly represented by their neighboring pixels. Low-rank representation methods [13], [14] decompose the entire HSI matrix into a low-rank background part and a sparse anomaly part. Tensor representation methods [15], [16], [17] regard hyperspectral data as a third-order tensor, which better preserves the spatial-spectral structure of the data [18]. Although representation-based methods have strong theoretical interpretability, they generally have the following problems: most require manual tuning of a large number of parameters, and flatten 3D hyperspectral data into a 2D matrix, destroying the inherent spatial-spectral structure information, limiting their performance improvement in complex scenes.

Benefiting from the powerful feature modeling capability of deep learning, deep neural network (DNN)-based methods have become the mainstream solution for current HAD. Unlike other methods, these methods do not require manual feature design or data distribution assumptions, and distinguish between background and anomalies through self-supervised reconstruction tasks [19], [20], [21]. In convolution-based anomaly detection methods, researchers mostly rely on the local feature extraction capability of convolutional neural networks (CNNs) to construct detection frameworks [22]. Autonomous HAD network based on fully convolutional autoencoder [23] realizes end-to-end HSI reconstruction and suppresses anomaly component reconstruction through adaptive weights. Weakly supervised discriminative learning with spectral-constrained generative adversarial network [24] enhances the difference in reconstruction errors between anomalies and the background. Deep low-rank prior (DeepLR) [25] incorporates low-rank prior constraints during the reconstruction process to improve background modeling accuracy. Such methods with fixed convolution kernels lack adaptive adjustment capabilities for multiscale targets and complex scenes, and fail to fully exploit discriminative information in the frequency domain [26]. This results in limited adaptability to spectral and structural variations.

The rise of graph neural networks (GNNs) provides a new technical path for HAD, which models feature correlations by constructing topological structures. Some methods [27], [28], [29] regard HSI pixels as graph nodes, construct static adjacency graphs based on spectral similarity, and capture global correlation features through graph convolution propagation. Other studies [30], [31] combine graph embedding with autoencoders to enhance the distinguishability of anomaly features. In addition, some derivative methods expand the technical boundary of DNN-based HAD, such as deep support vector data description [32] which transforms anomaly detection into a hypersphere classification problem in the feature space, and low-rank embedded network [33] which extracts deep spectral features through joint training of an autoencoder and a Gaussian mixture model.

In summary, although existing methods have achieved significant progress, they still suffer from three major limitations. First, frequency-domain feature modeling remains insufficient, as conventional transforms lack adaptability to spectral variations and fail to fully exploit discriminative phase and amplitude information. Second, spatial multiscale modeling is largely static, with fixed convolution kernels and neighborhood definitions that are inadequate for handling targets of diverse scales under complex background conditions. Third, spectral–spatial interaction mechanisms are underdeveloped, as most approaches rely on simplistic fusion strategies and lack effective multiscale feature propagation and structured regularization, limiting the exploitation of global–local topological dependencies.

To address the above problems, this paper proposes a HAD method learnable frequency domain phase residual and graph-contrastive dynamic convolution (LFGDC) network. We move beyond fixed frequency-domain transforms and realize adaptive transformation in the frequency spectrum domain to enhance spectral robustness, optimize band weights through attention mechanism, and improve target representation capability through multiscale fusion. The core contributions of this study are as follows:

1) A learnable frequency-domain feature modeling mechanism is introduced. By embedding learnable frequency domain phase residual block (LFPRB), adaptive rotation modulation of frequency-domain phase is realized, combined with a multiscale spectral enhancement strategy to extract complementary information at different frequency scales. In addition, a lightweight spectral channel gating module is used to realize adaptive weighting of spectral channels and enhancement of significant spectral bands, thereby significantly improving the frequency-domain discriminability and expression stability of spectral features.
2) A multiscale adaptive dynamic convolution module (MDCM) is proposed. This structure generates dynamic convolution kernels driven by input features to realize input-dependent adjustment of convolution weights, and extracts feature representations of different receptive fields under multiple convolution branches. This design enables the network to have spatial multiscale response and context adaptive capabilities, enhancing the modeling performance for complex scenes and target structure changes.
3) A graph-contrastive structure fusion module (GCSFM) is constructed. By adaptively constructing multiscale adjacency graphs based on similarity in the feature space and performing graph convolution propagation under different neighborhood scales, the model can capture local and global topological correlation features. At the same time, contrastive learning loss is introduced to aggregate intra-class features and separate inter-class features, enhancing feature discriminative capability from the structural level.

The rest of this paper is organized as follows. Section II provides an overview of related works on HAD, Section III elaborates on the proposed method. In Section IV, we present experimental results and analysis, comparing the performance of our framework with other methods. Finally, Section V gives concluding remarks.

II. Related Work

This section briefly reviews the related work of traditional methods, and deep learning-based methods, respectively.

A. Traditional Methods

Some subspace representation methods assume that background data lies in a low-dimensional subspace, while anomaly pixels deviate from this subspace, and realize anomaly detection by learning the background subspace and calculating the projection error of pixels to the subspace. Sparse representation methods assume that anomaly pixels have sparse representation characteristics under the background dictionary, and separate the background and anomalies by solving sparse coefficients. Mairal et al. propose a dictionary learning-based sparse representation [34] method, which uses redundant information of the background to improve representation accuracy. The collaborative representation detection [35] method constructs a dictionary with local neighboring pixels to improve computational efficiency.

Li et al. [36] builds a new set of HAD benchmark datasets for improving the robustness of the algorithm in complex scenarios. Sun et al. propose the low-rank and total variation-based anomaly detection (LARTVAD) [37] method, which operates directly on third-order tensors, uses tensor singular value decomposition to characterize the global low-rankness of the background, and combines total variation regularization to constrain spatial smoothness. Representation-based methods usually rely on fixed linear transforms (such as Fourier transform) or predefined dictionaries, lacking adaptive learning capability for discriminative frequency-domain features, and spatial feature extraction is often static and fixed, making it difficult to adapt to variable target scales and morphologies.

B. Deep Learning-Based Methods

With the breakthrough progress of deep learning technology in feature extraction and representation learning, deep learning-based HAD methods have become the current research mainstream, mainly covering five core architectures: autoencoders, generative adversarial networks, convolutional neural networks, transformers, and graph neural networks.

AE-based methods are the most widely used frameworks. The core idea is to use the reconstruction capability of AE to learn background features and realize discrimination through the high reconstruction error of anomaly pixels [38], [39], [40]. Early stacked AEs focuse on the spectral dimension and ignore spatial information, resulting in poor detection results for complex structural anomalies. Convolutional AEs introduce convolution operations to extract spatial-spectral joint features, but fixed-size convolution kernels are difficult to adapt to multiscale targets and have insufficient robustness to spectral variation. Subsequent improved adversarial AEs [41] constrain feature distribution through an adversarial mechanism but are prone to mode collapse. Variational AEs [42] improve generalization capability but have insufficient distinguishability for weak anomaly targets, cascaded AEs and pixel-level attention AEs enhance feature extraction but have problems such as parameter redundancy and lack of global guidance.

CNN-based methods construct end-to-end models with the help of the spatial feature extraction capability of CNNs. Liu et al. propose a multiscale CNN method [43], which uses multi-size convolution kernels to extract multiscale features. Multi-frequency transform guides dual-path CNN realizes anomaly enhancement and background restoration, but the decomposition threshold is fixed. Wang et al. [44] propose a multiscale fusion network with global weighting, integrating spectral–spatial features via 3-D weighting to suppress heterogeneous interference.

Transformer-based methods capture long-distance dependencies through self-attention mechanisms, effectively modeling global spectral correlations and long-distance spatial dependencies. The spectral–spatial transformer dual-branch fuses spectral and spatial features [45], but the fusion method is simple and inefficient. The lightweight spatial-window transformer reduces complexity by limiting attention windows, the hybrid transformer-CNN architecture combines the local feature extraction of CNNs with the global dependency modeling of transformers [46], retaining global information while improving efficiency [47].

GNN-based methods mine topological correlations by constructing pixel adjacency graphs [48], adapting to non-Euclidean structure data [49]. A method based on graph pixel selection achieves anomaly detection by solving a quadratic programming problem on a vertex- and edge-weighted graph in [50]. Spectral graph convolutional networks rely on fixed similarity thresholds to construct adjacencies. It [51] fuses dual-dimensional correlations but has a fixed weight fusion ratio. Liu et al. [52] captures local and global topological features but uses simple concatenation for fusion, and fixes neighborhood scales cannot adapt to dynamic targets.

III. Methodology

Let $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$ denote a HSI, where H, W, and B represent the spatial height, width, and the number of spectral bands, respectively. In this work, we adopt a reconstruction-based paradigm. A neural network $\mathcal{F}(\cdot;\Theta)$ is trained to model background structures, while anomalous pixels are expected to produce larger reconstruction errors, where Θ denotes the set of all learnable network parameters. The anomaly score at spatial location (i, j) is defined as:

$$R(i,j) = \left\|\mathbf{X}(i,j) - \hat{\mathbf{X}}(i,j)\right\|_2^2, \tag{1}$$

where $\hat{\mathbf{X}} = \mathcal{F}(\mathbf{X};\Theta)$ denotes the reconstructed HSI.

The overall architecture of the proposed method is illustrated in Fig. 1. The network consists of three major components: LFPRB for extraction of spectral discriminant features, MDCM for spatial feature extraction, and GCSFM for spectral–spatial consistency enhancement.

The proposed network follows a hierarchical modeling strategy. Specifically, spectral modeling is first conducted in the frequency domain to capture intrinsic spectral characteristics. Subsequently, spatial feature interactions are enhanced via

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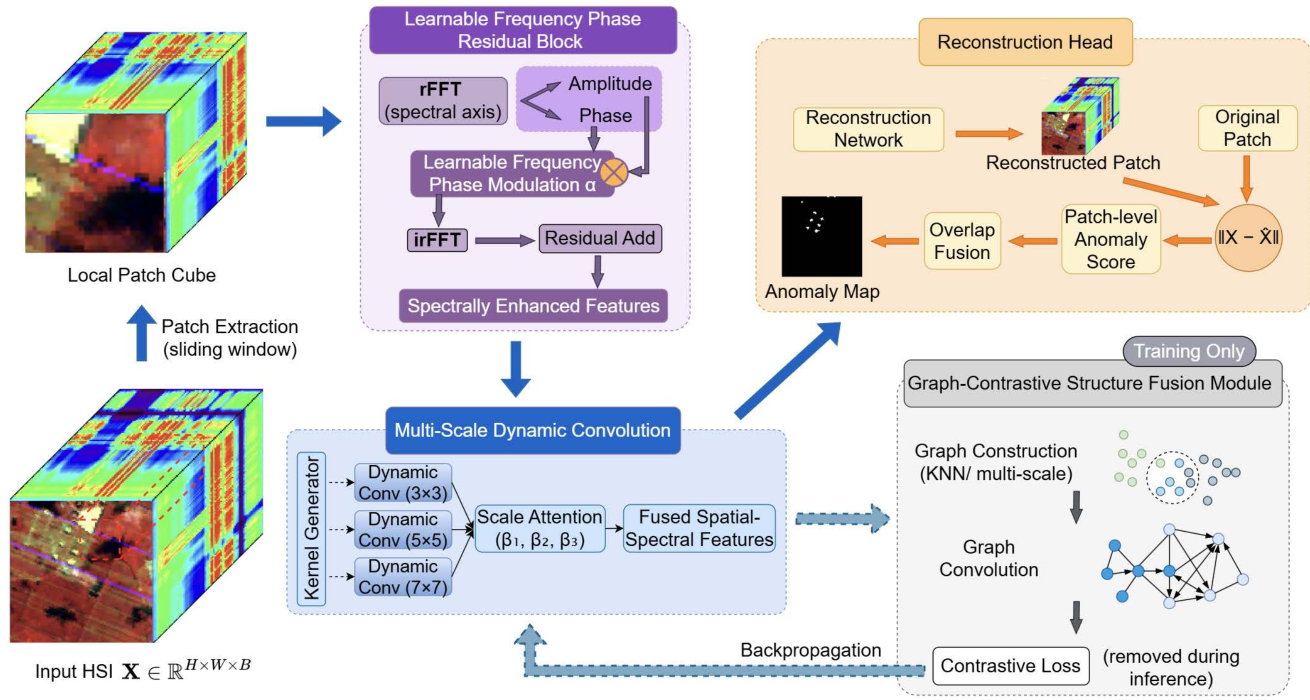


Fig. 1: Overall framework of the proposed HAD method. LFPRB performs frequency-domain spectral enhancement, followed by MDCM for input-adaptive multiscale spatial modeling. In parallel, GCSFM is imposed on intermediate features during training to enforce global–local structural consistency via graph-contrastive learning, without introducing additional inference overhead.

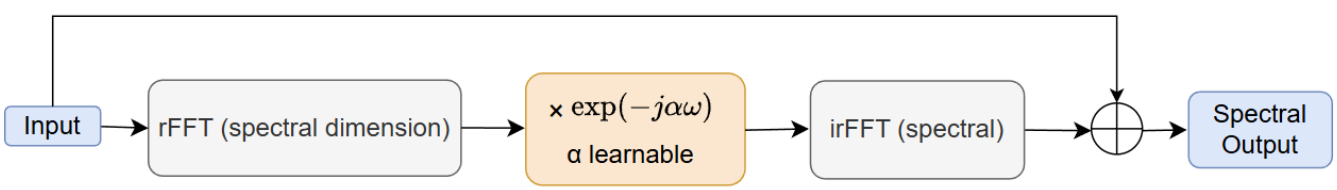


Fig. 2: Structure of the LFPRB. A learnable frequency domain phase modulation is applied to spectral frequency components, followed by a residual connection to preserve the original spectral information.

multiscale dynamic convolution. Finally, structural consistency is imposed through graph-based contrastive learning during training to further regularize background representations.

A. LFPRB

Spectral information plays a fundamental role in HAD. However, conventional convolutional networks mainly operate in the spatial domain and implicitly model spectral correlations through stacked convolutions, which limits their ability to explicitly capture discriminative spectral frequency characteristics. Moreover, fixed spectral transforms lack adaptability to spectral variability caused by illumination changes, material heterogeneity, and complex background distributions.

To explicitly exploit spectral frequency-domain characteristics while maintaining adaptivity, we introduce a LFPRB to enhance spectral discrimination, as shown in Fig. 2. Given an input hyperspectral feature tensor $\mathbf{X} \in \mathbb{R}^{B\times H\times W}$, a real-valued Fourier transform is applied along the spectral dimension:

$$\mathbf{X}_f = \mathrm{rFFT}(\mathbf{X}), \tag{2}$$

where $\mathbf{X}_f \in \mathbb{C}^{(\lfloor B/2\rfloor+1)\times H\times W}$. Instead of adopting a fixed spectral transform, a learnable frequency domain phase modulation is performed:

$$\mathbf{X}'_f = \mathbf{X}_f \odot \exp(-j\alpha\omega), \tag{3}$$

where $j = \sqrt{-1}$ denotes the imaginary unit, which is distinct from the spatial index j used in (i, j), ω denotes the normalized spectral frequency indices, and α is a learnable frequency domain parameter controlling the degree of phase rotation. This operation enables adaptive adjustment of spectral phase information, allowing the network to emphasize discriminative frequency components and suppressing background-dominated spectral responses in a data-driven manner.

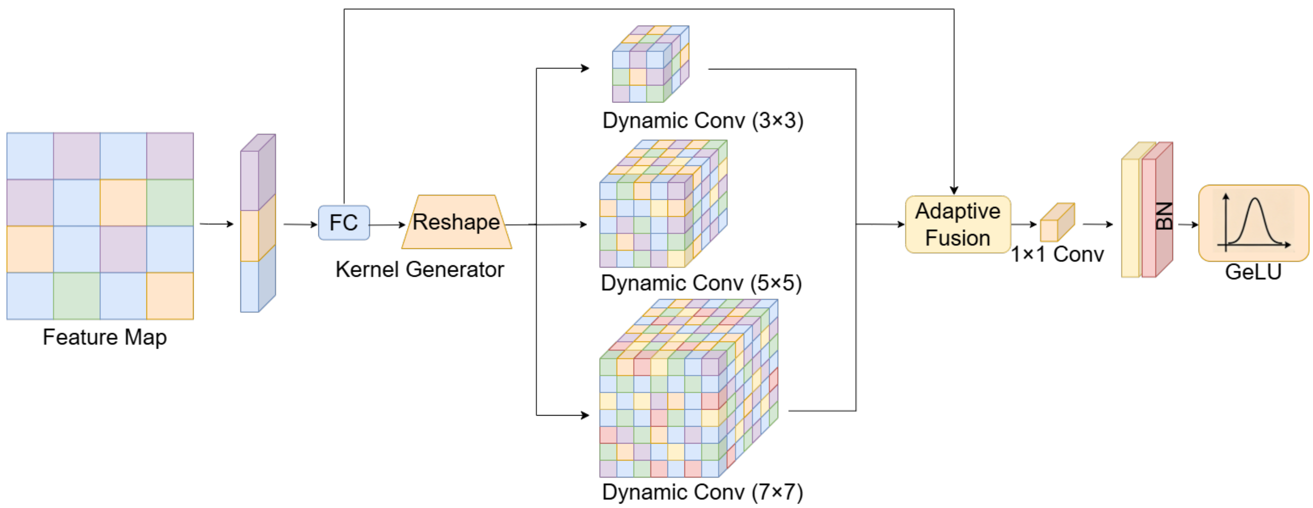


Fig. 3: Structure of the MDCM. Dynamic convolution kernels with different receptive fields are generated adaptively according to the input features for flexible spatial modeling.

The modulated frequency-domain features are then transformed back to the spectral domain using the inverse real FFT:

$$\tilde{\mathbf{X}} = \mathrm{irFFT}(\mathbf{X}'_f), \tag{4}$$

which results in a spectrally enhanced representation. To preserve the original spectral information and stabilize network optimization, a residual connection is introduced:

$$\mathbf{X}_{\mathrm{LFPRB}} = \tilde{\mathbf{X}} + \mathbf{X}. \tag{5}$$

Through learnable frequency domain phase modulation, LFPRB enhances spectral discriminability while retaining background stability, providing a robust spectral foundation for subsequent spatial modeling.

B. MDCM

Hyperspectral anomaly targets often exhibit irregular spatial structures and diverse spatial scales. Fixed convolution kernels with limited receptive fields are insufficient to effectively model such variability, especially under complex and heterogeneous background conditions. Although multiscale convolutional architectures can partially alleviate this issue, their convolution kernels remain static and lack adaptability to local feature distributions. To address these limitations, we propose a MDCM that performs input-dependent spatial feature extraction by dynamically adjusting convolutional kernels across multiple receptive fields, as shown in Fig. 3.

Given an intermediate feature map $\mathbf{F} \in \mathbb{R}^{C\times H\times W}$, the feature tensor is rearranged to $C \times H \times W$ for convolution operations, where C denotes the number of feature channels, and MDCM consists of multiple parallel convolution branches with different kernel sizes, each corresponding to a distinct receptive field. Instead of using fixed convolution weights, a lightweight kernel generation mechanism conditions the convolution kernels on the input feature representation, enabling adaptive modulation of spatial filters according to local spectral–spatial characteristics. Specifically, the dynamic convolution kernels are generated through a lightweight kernel generation network. For each branch, the input feature map is first globally aggregated and then fed into a sequence of fully connected (FC) layers, followed by batch normalization (BN) and Gaussian Error Linear Unit (GeLU) activation, to produce the scale-specific convolution weights.

For the s-th convolution branch, the dynamic convolution operation can be expressed as:

$$\mathbf{F}_s = \mathrm{Conv}(\mathbf{F}, \mathbf{W}_s(\mathbf{F})), \tag{6}$$

where $\mathbf{Conv}_s(\cdot)$ denotes a standard convolution operation, $\mathbf{W}_s(\cdot)$ denotes the dynamically generated convolution kernel for s-th scale, and $\mathbf{W}_s(\mathbf{F})$ is generated by a lightweight parameter prediction network composed of FC, BN, and GeLU layers. The use of FC layers enables flexible cross-channel interaction, BN stabilizes the dynamic kernel learning process, and GeLU introduces non-linearity to enhance the expressive capacity of the generated kernels, while keeping the overall computational overhead low. This formulation allows the receptive field response to adapt to spatial context variations, enhancing sensitivity to anomalies of different sizes and shapes.

The outputs of all branches are subsequently fused through an attention-based weighting mechanism:

$$\mathbf{F}_{\mathrm{MDCM}} = \sum_{s=1}^{N} \beta_s \mathbf{F}_s, \quad \sum_{s=1}^{S} \beta_s = 1, \tag{7}$$

where β_s represents the adaptive importance weight of the s-th scale, inferred from the input features. Specifically, the scale weights are estimated via a lightweight attention mechanism, in which the multi-scale branch outputs are first summarized by global average pooling and then passed through a small fully connected network to produce scale-wise responses. A softmax normalization is subsequently applied to obtain β_s, ensuring that $\sum_s \beta_s = 1$. This adaptive fusion strategy enables the network to emphasize the most informative spatial scales while suppressing redundant or background-dominated responses.

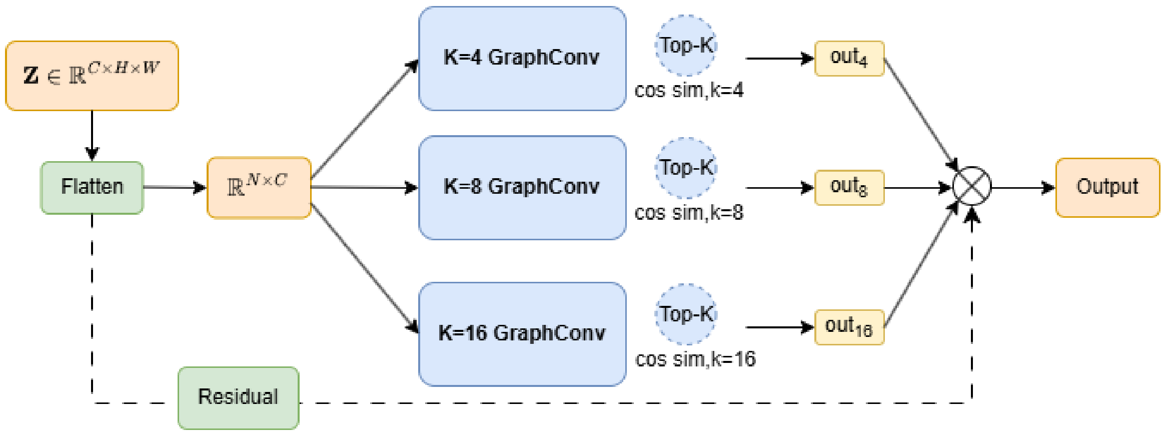


Fig. 4: Structure of the GCSFM. Multiscale graph convolution and contrastive learning are employed to enhance global local spectral–spatial consistency.

By jointly leveraging multiscale receptive fields and input-dependent convolutional kernels, MDCM provides flexible and context-aware spatial modeling, which is particularly beneficial for HAD in scenes with complex spatial layouts and scale-varying targets.

C. GCSFM

Although reconstruction-based HAD methods can effectively model background distributions, they often suffer from insufficient feature compactness in the latent space, which leads to background dispersion and blurred separation between background and anomaly representations. This issue is particularly prominent in complex scenes with heterogeneous spatial structures and spectral variability.

To further enhance spectral–spatial consistency and improve feature discriminability during training, we introduce a GCSFM, which incorporates structured neighborhood modeling and contrastive constraints into the representation learning process, as shown in Fig. 4.

Given the intermediate feature representation $\mathbf{Z} \in \mathbb{R}^{C\times H\times W}$, each spatial location is treated as a graph node. To capture local spectral–spatial correlations, a similarity graph is constructed based on cosine similarity in the feature space. For each node n, its neighborhood $\mathcal{N}_K(n)$ is defined as the set of its top-K nearest neighbors, and the corresponding adjacency matrix is formulated as:

$$A_{nm} = \begin{cases} 1, & \text{if } m \in \mathcal{N}_K(n), \\ 0, & \text{otherwise.} \end{cases} \tag{8}$$

Note that the resulting adjacency matrix is generally asymmetric due to the directed KNN construction. Multiscale graphs are constructed by varying the neighborhood size K, enabling both local and global structural modeling. Based on the constructed graph, graph convolution is performed to aggregate neighborhood information:

$$\mathbf{H} = \sigma(\mathbf{AZW}), \tag{9}$$

where $\mathbf{W}$ is a learnable weight matrix and $\sigma(\cdot)$ denotes a nonlinear activation function. This operation enables each node to integrate contextual information from spectrally and spatially similar neighbors, reinforcing background consistency while preserving local structural variations.

Let $\mathbf{h}_n$ denote the graph-enhanced feature representation of the n-th node in $\mathbf{H}$. To further regularize the learned representations, a contrastive learning objective is imposed on these graph-enhanced features. Specifically, for each anchor feature $\mathbf{h}_n$, a positive $\mathbf{h}_n^+$ is constructed by pairing features of the same node under different graph scales or feature perturbations, while features from other nodes serve as negative samples. The contrastive loss is defined as:

$$\mathcal{L}_{\text{con}} = -\sum_n \log \frac{\exp(\text{sim}(\mathbf{h}_n, \mathbf{h}_n^+)/\tau)}{\sum_m \exp(\text{sim}(\mathbf{h}_n, \mathbf{h}_m)/\tau)}, \tag{10}$$

where $\text{sim}(\cdot,\cdot)$ denotes the cosine similarity function and τ is the temperature parameter This loss encourages compact background representations while enlarging the embedding gap between background and anomalous pixels.

By jointly leveraging graph-based neighborhood aggregation and contrastive regularization, GCSFM encourages compact background representations while enlarging the embedding separation between background and anomalous pixels. It is worth noting that GCSFM is only activated during training and is completely removed during inference, introducing no additional computational overhead at test time.

The overall training objective is formulated as:

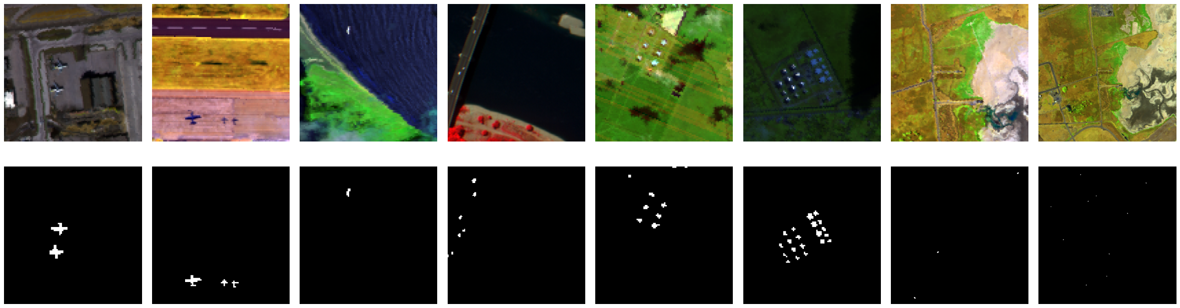
$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda \mathcal{L}_{\text{con}}, \tag{11}$$

where $\mathcal{L}_{\text{rec}}$ denotes the reconstruction loss and λ controls the contribution of the contrastive term.

During inference, only the reconstruction branch is retained. The anomaly map is generated based on the pixel-wise reconstruction residual defined in Eq.(1) followed by spatial smoothing to suppress isolated noise responses.

IV. Experimental Results

In this section, eight real hyperspectral datasets are used to evaluate the performance of the proposed HAD algorithm through experiments. Detailed descriptions of these real datasets are provided below. All experiments are conducted in



(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 5: The pseudo-color images and ground truth maps of the eight HSI datasets. (a) Los Angeles, (b) Gulfport, (c) Cat Island, (d) Pavia, (e) Texas Coast, (f) Gainesville, (g) Wasteland-WZW, (h) Wasteland-DDC.

the Python 3.11 environment on a personal computer equipped with an AMD Ryzen 9 8940HX CPU, an NVIDIA RTX 5060 GPU, 16 GB memory, and 64-bit Windows 11 operating system. In terms of deep learning frameworks, PyTorch 2.9.0 is used.

A. Datasets

The first dataset is Los Angeles, captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over the region of Los Angeles. After removing water absorption and low signal-to-noise ratio bands, 205 spectral bands are retained, with a spatial size of 100×100 pixels. The second dataset, Gulfport, is also an airport area captured by the airborne AVIRIS sensor. It has a size of 100×100 pixels and retains 191 bands. The third dataset, Cat Island, is a beach region also captured by the AVIRIS sensor, containing 100×100 pixels and 188 spectral bands. The fourth dataset, Pavia, is a city acquired by the reflective optics system imaging spectrometer (ROSIS) sensor, with a spatial dimension of 150×150 pixels and 102 spectral bands. The fifth dataset is Texas Coast, which consists of images of an urban area collected by the airborne AVIRIS sensor. The spatial area covered by the dataset is 100×100 pixels, and 204 spectral bands are retained removing noise bands. The sixth dataset is Gainesville, urban area images also collected by the airborne AVIRIS sensor. The dataset covers a spatial area of 100×100 pixels, with 207 spectral bands retained after noise band removal, respectively. The seventh dataset, Wasteland-WZW, which is collected from a specific wasteland area, contains three camouflage. These are made of a special synthetic material designed to blend with the natural environment. The areas corresponding to these camouflage in the image have 21 anomalous pixels. The dataset has a size of 256×256 pixels and 198 spectral bands. The final dataset, Wasteland-DDC, is characterized by a spatial dimension of 512×512 pixels. This dataset features a total of ten missile - vehicle targets, and after excluding noisy bands, 198 spectral bands are preserved, containing a total of 64 anomalous pixels to be detected. These datasets feature diverse types of anomaly targets, and their pseudo-color images along with ground truth maps are shown in Fig. 5.

B. Comparative Algorithms and Evaluation Metrics

1) Comparative Algorithms: To comprehensively demonstrate the superiority of our proposed method, we selected seven state-of-the-art HAD algorithms for comparison, including low-rank and total variation-based anomaly detection (LARTVAD) [37], transformer-based AE framework (TAEF) [53], low-rank and sparse matrix decomposition-based Mahalanobis distance anomaly detection (LSMAD) [54], kernel isolation forest detection (KIFD) [55], deep low-rank prior (DeepLR) [25], gated transformer for HAD (GT-HAD) [56], and background suppression diffusion model (BSDM) [57]. The LSMAD algorithm measures the degree of anomaly by calculating the Mahalanobis distance between the pixel under test and the statistical characteristics of its local background. LARTVAD is a tensor-based method that exploits the low-rank property and piecewise smoothness of the background tensor [58]. TAEF employs the attention mechanism of transformers to handle long-range dependencies and captures global information. KIFD is an unsupervised detection algorithm based on kernel density estimation. Deep-LR integrates model-driven low-rank priors into a data-driven network and imposes low-rank constraints on the network outputs. GT-HAD utilizes the spatial-spectral similarity of HSI to guide the reconstruction process.

2) Evaluation Metrics: Several classical and commonly used metrics are employed to evaluate the performance of these algorithms. First, the receiver operating characteristic (ROC) curve is employed to evaluate anomaly detection performance by illustrating the trade-off between the true positive rate (TPR) and the false positive rate (FPR) under different detection thresholds. Specifically, TPR measures the proportion of abnormal pixels that are correctly detected and is defined as the ratio of true positives (TP) to the total number of abnormal pixels, where TP denotes the number of abnormal pixels correctly classified as anomalies and false negatives (FN) denote the number of abnormal pixels incorrectly classified as background. FPR measures the proportion of background pixels that are incorrectly classified as anomalies and is defined as the ratio of false positives (FP) to the total number of background pixels, where FP denotes the number of background pixels incorrectly classified as anomalies and true negatives (TN) denote the number of

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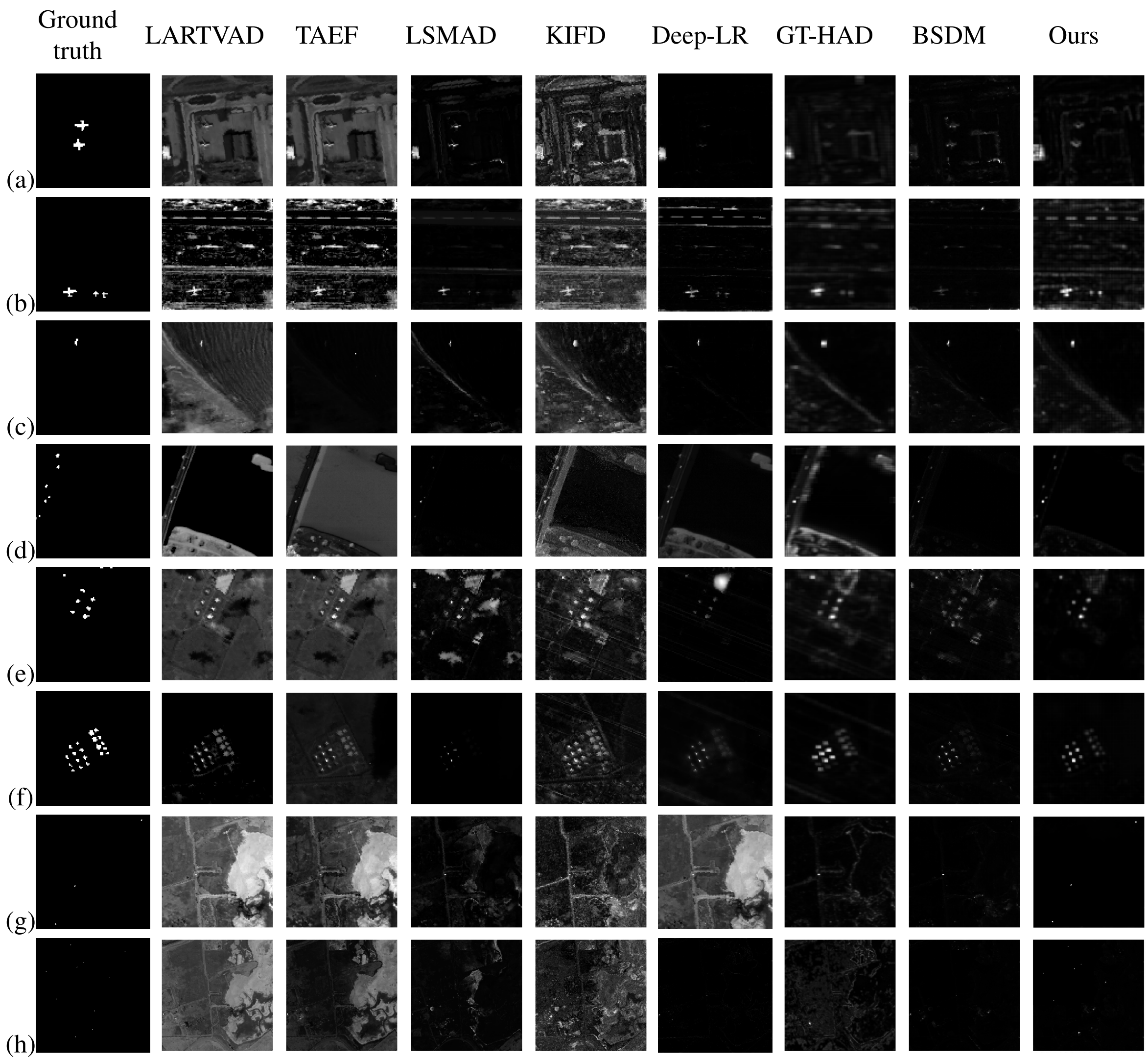

Fig. 6: The detection maps of different methods on eight datasets. (a) Los Angeles, (b) Gulfport, (c) Cat Island, (d) Pavia, (e) Texas Coast, (f) Gainesville, (g) Wasteland-WZW, (h) Wasteland-DDC. All anomaly maps are normalized to the range [0, 1] for visualization purposes, where brighter intensities indicate higher anomaly scores.

background pixels correctly classified as background. Second, the area under the ROC curve (AUC) is used to measure the overall performance of the algorithm, the closer the AUC is to 1, the better the average performance. Finally, the background-separated anomaly map is drawn through a box-plot, which clearly shows the distribution difference of the anomaly scores output by the algorithm on real background pixels and real abnormal pixels, thereby intuitively evaluating the separation ability of the algorithm. An anomaly box that is overall much higher than the background box and has a good separation between the two boxes indicates that the anomaly detection algorithm has excellent performance.

C. Detection Performance

The detection maps of different methods on eight HSI datasets are shown in Fig. 6. As can be seen from the figure, the proposed method can achieve a balance between anomaly detection and background suppression in all scenarios. The LARTVAD, TAEF, and KIFD methods are susceptible to background texture interference, resulting in numerous false positive annotations. In contrast, our method can successfully eliminate texture interference and accurately locate anomaly targets. Moreover, compared with the deep learning methods of Deep-LR and GT-HAD, the boundaries of anomaly targets in our method are clearer, with no obvious target diffusion phenomenon. In the camouflage net detection task of Wasteland-WZW, the proposed method can effectively overcome the spectral confusion between the camouflage net and the background. However, due to the insufficient capture of low-frequency spectral differences, LSMAD and BSDM miss the detection of some anomaly pixels. These detection maps strongly demonstrate that the proposed method can effectively highlight anomaly targets while significantly suppressing background noise.

Fig. 7 shows the box-plot results of each method. The green boxes represent the distribution of anomaly scores for

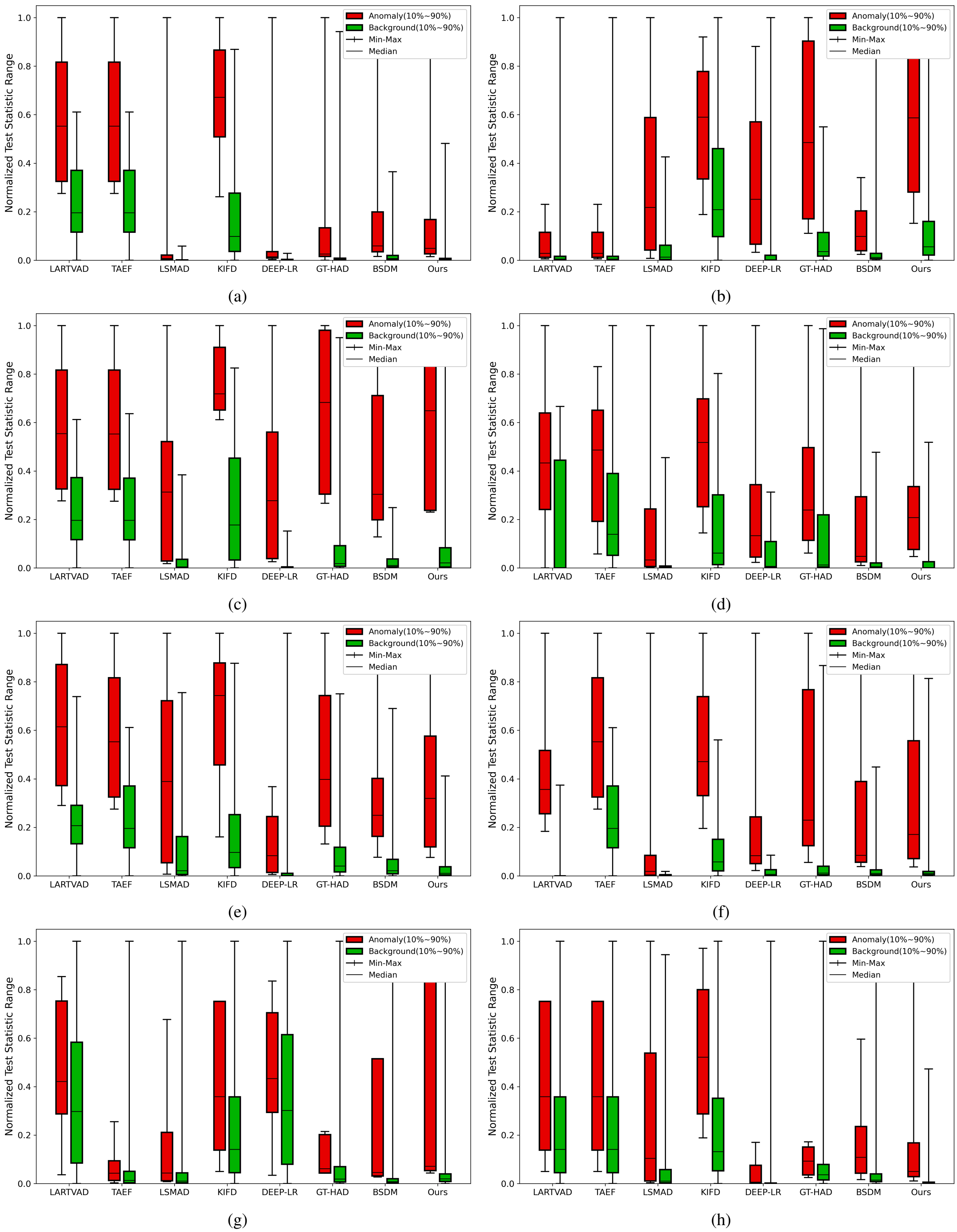

Fig. 7: Box-plots of the compared methods on the eight datasets. (a) Los Angeles, (b) Gulfport, (c) Cat Island, (d) Pavia, (e) Texas Coast, (f) Gainesville, (g) Wasteland-WZW, (h) Wasteland-DDC.

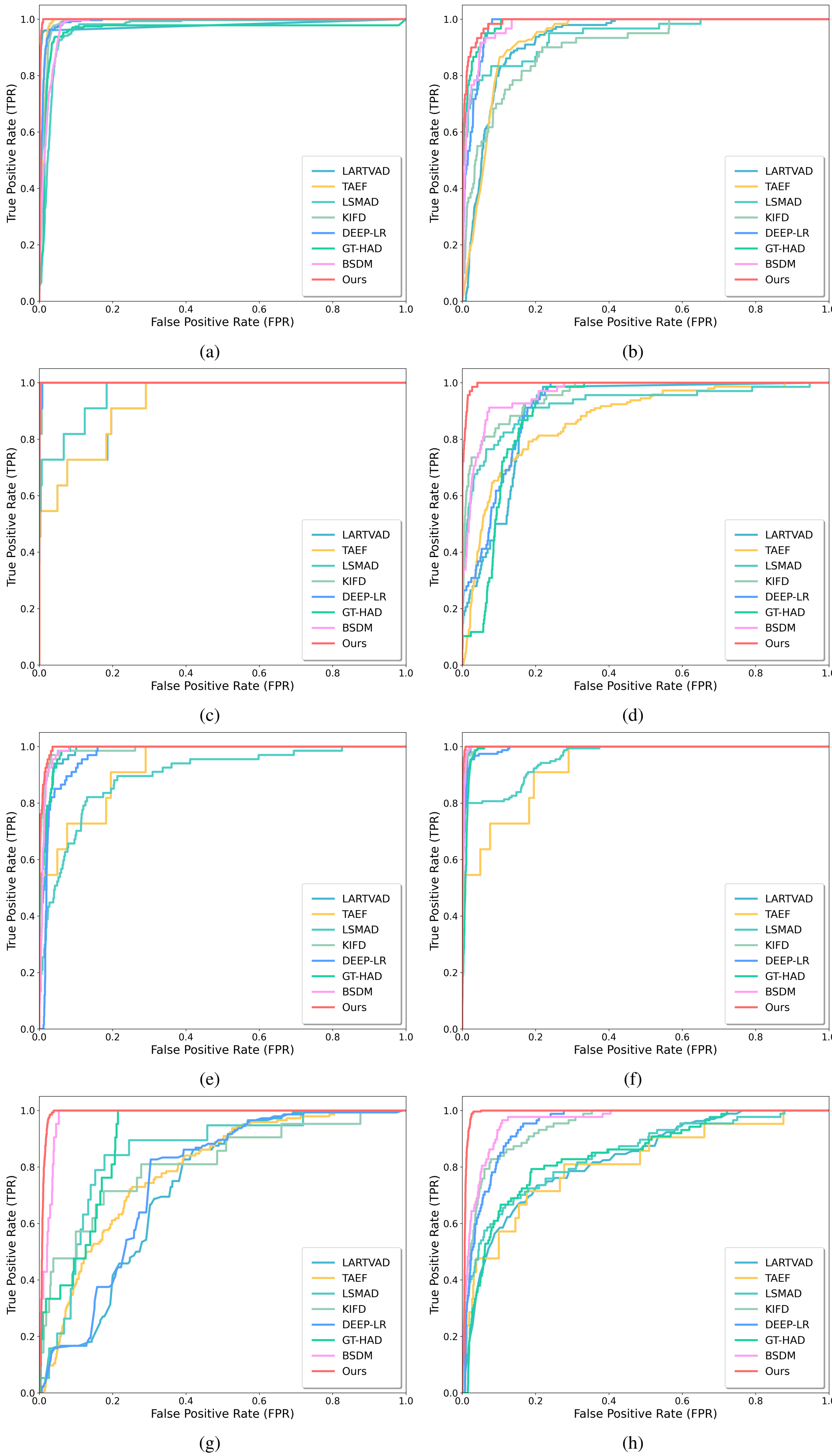

Fig. 8: ROC curves of the different methods on the eight datasets. (a) Los Angeles, (b) Gulfport, (c) Cat Island, (d) Pavia, (e) Texas Coast, (f) Gainesville, (g) Wasteland-WZW, (h) Wasteland-DDC.

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TABLE I: AUC Values of the Different Methods on the Eight Hyperspectral Datasets

Dataset	LARTVAD	TAEF	LSMAD	KIFD	Deep-LR	GT-HAD	BSDM	Ours
Los Angeles	0.9781	0.9910	0.9670	0.9887	0.9911	0.9603	0.9807	**0.9982**
Gulfport	0.9259	0.9304	0.9410	0.9017	0.9772	0.9862	0.9783	**0.9902**
Cat Island	0.9271	0.9271	0.9648	0.9985	0.9990	0.9990	0.9996	**0.9996**
Pavia	0.9003	0.8696	0.9218	0.9568	0.9178	0.8992	0.9627	**0.9965**
Texas Coast	0.9834	0.9276	0.8973	0.9903	0.9674	0.9866	0.9882	**0.9959**
Gainesville	0.9963	0.9276	0.9601	0.9972	0.9931	0.9904	0.9968	**0.9994**
Wasteland-WZW	0.7320	0.7899	0.8479	0.8130	0.7557	0.8932	0.9789	**0.9910**
Wasteland-DDC	0.8294	0.8130	0.8462	0.9458	0.9478	0.8454	0.9623	**0.9921**

TABLE II: Execution Time of the Different Algorithms on the Eight Datasets (Unit: Seconds)

Dataset	LARTVAD	TAEF	LSMAD	KIFD	Deep-LR	GT-HAD	BSDM	Ours
Los Angeles	228.75	60.51	1.84	0.63	86.57	216.94	16.53	280.34
Gulfport	231.79	95.30	1.87	0.61	87.24	215.35	15.27	282.51
Cat Island	236.83	58.25	1.83	0.62	85.31	331.15	16.79	291.06
Pavia	314.26	63.26	2.98	0.70	97.63	356.47	37.25	343.96
Texas Coast	265.44	92.48	1.84	1.02	102.98	232.77	29.09	305.78
Gainesville	264.92	88.56	1.91	0.64	92.17	222.06	29.29	304.92
Wasteland-WZW	381.51	243.12	8.90	1.54	1934.57	2443.39	150.31	1921.38
Wasteland-DDC	839.77	752.27	40.51	2.79	7548.32	8773.56	1580.99	7957.94

background pixels, and the red boxes represent the score distribution of anomaly pixels. We can observe that our method can achieve excellent background suppression and separation between background and anomalies on all eight datasets. On the Wasteland-DDC dataset, the median score of the anomaly box is more than three times higher than that of the background box, far outperforming other comparison algorithms. The scores of the background boxes of LARTVAD and LSMAD fluctuate significantly, indicating their weak adaptability to complex backgrounds. In contrast, the scores of the background box of the proposed method are concentrated in the low-score range, and the box is compact, suggesting stable background suppression. Although GT-HAD and BSDM are superior to other algorithms in background suppression, GT-HAD fails to effectively separate the background from anomalies, and the anomaly scores of BSDM are relatively low. In complex urban scenarios such as Texas Coast and Gainesville, the inter-quartile range of the anomaly box of the proposed method is smaller, indicating higher consistency in the scores of anomaly targets and less influence from target scale changes. Overall, the proposed method can effectively suppress the background and separate anomalies.

The ROC curves of different methods are shown in Fig. 8, and the AUC scores are listed in Table I, where a larger value indicates better performance, and the best results are highlighted in bold. The ROC curve of the proposed method is closer to the upper-left corner of the coordinate axis on all datasets, demonstrating superior detection trade-off performance. On the Los Angeles dataset, the TPR of the proposed method reaches 0.92 when FPR = 0.01, while the TPRs of Deep-LR and GT-HAD are only 0.78 and 0.83 respectively, indicating that the proposed method has stronger anomaly detection ability at a low false positive rate. On the Pavia dataset, due to the spectral heterogeneity of the background, the ROC curves of KIFD and TAEF fluctuate significantly, while the curve of the proposed method is smooth and always above those of other algorithms, reflecting stronger scene adaptability. By observing Table I, it is evident that the AUC score of the proposed method is superior to all other methods on all datasets, with an average AUC value of 0.987, significantly higher than other comparison algorithms. In summary, our method better balances the relationship between the detection rate and the false alarm rate, consistently achieving better AUC scores across all datasets.

Table II presents the computational costs of the different algorithms on the eight test datasets. The KIFD algorithm takes the least time on all test datasets. The GT-HAD algorithm has the highest computational cost. The proposed method exhibits increased inference time when processing large-scale HSI. This behavior is mainly attributed to the patch-wise sliding-window inference strategy adopted in our implementation, which is commonly used in reconstruction-based HAD to ensure local spectral–spatial consistency. It should be noted that the reported runtime reflects the cumulative cost of densely scanning overlapping patches over large images, rather than the intrinsic complexity of the proposed network.

Moreover, the GCSFM is exclusively employed during training and is completely removed during inference, which further ensures that the test-time complexity remains independent of graph construction. Therefore, the increased runtime on large images primarily arises from implementation-level design choices rather than algorithmic inefficiency. In practical applications, the proposed method can be efficiently accelerated through larger patch strides, parallel processing, or fully convolutional inference, without affecting the learned model parameters.

In conclusion, our proposed method demonstrates excellent performance in terms of detection maps, box-plots, ROC curves, and AUC scores. Despite a moderate parameter scale (4.22M), the proposed method achieves state-of-the-art performance, demonstrating a favorable balance between model complexity and detection accuracy.

D. Ablation Study

To quantitatively analyze the contribution of each proposed component, we conduct an ablation study on two representative datasets, Texas Coast and Pavia, which contain complex background structures and scale-varying anomaly targets. All variants are trained under the same settings to ensure fair comparison. Four variants are implemented as follows:

- **Baseline**: A reconstruction-based network consisting of a lightweight convolutional encoder–decoder with residual connections, without LFPRB, MDCM, or GCSFM.
- **+LFPRB**: The baseline model augmented with the proposed LFPRB to enable adaptive frequency-domain spectral modeling.
- **+MDCM**: Further introducing the MDCM for adaptive spatial feature extraction with input-dependent convolution kernels.
- **Full (LFPRB + MDCM + GCSFM)**: The complete proposed model, which additionally incorporates the GCSFM during training to enhance spectral–spatial consistency via graph-based contrastive regularization.

1) Visual Comparison: Fig. 9 presents the visual anomaly detection results of different ablation variants on the Pavia dataset.

The Baseline model produces noticeable false alarms in homogeneous background regions, indicating limited discriminability when relying solely on the reconstruction loss. After introducing LFPRB, spectral discrimination is improved, and some background interference is suppressed due to adaptive frequency-domain phase modulation. With the addition of MDCM, anomaly targets become more spatially compact and better localized, benefiting from multi-scale and input-adaptive spatial modeling.

The full model, which incorporates GCSFM during training, achieves the cleanest anomaly maps with the sharpest target boundaries and minimal background responses. This demonstrates that graph-based contrastive regularization effectively enforces background compactness in the feature space and enlarges the separation between background and anomalous representations.

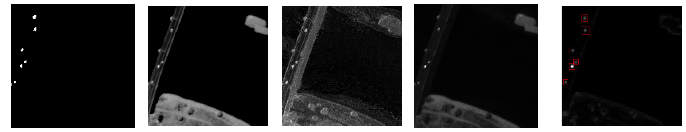

Fig. 9: From left to right: Ground-truth, Baseline, +LFPRB, +MDCM, The full model on Pavia.

TABLE III: AUC and GPU inference time (ms) of ablated variants on Pavia (Best results are in boldface)

Model	AUC ↑	Time (ms)
Baseline	0.9492	21
+LFPRB	0.9793	23
+MDCM	0.9897	25
the full model	**0.9974**	27

2) Quantitative Evaluation: Table III summarizes the AUC values and inference time on the Pavia dataset for the different ablation variants.

Introducing LFPRB improves the AUC by effectively exploiting discriminative spectral frequency information. The further inclusion of MDCM brings an additional performance gain by enhancing spatial adaptability to targets of different scales and shapes. The full model achieves the highest AUC, confirming that GCSFM provides complementary benefits by regularizing feature representations through graph-based contrastive learning during training.

Although the inference time slightly increases with the addition of LFPRB and MDCM, the overall computational overhead remains limited. This increase mainly arises from spectral frequency modulation and multi-branch convolution operations, while no graph construction or contrastive computation is involved at inference time.

3) Parameter Sensitivity: We additionally investigate three key hyper-parameters on Texas Coast.

TABLE IV: Impact of α on Texas Coast (Best results are in boldface)

α	0.1	0.5	1.0
AUC	0.9891	0.9922	**0.9840**

The α is the LFPRB parameter. This parameter controls the degree of spectral phase rotation in the frequency domain and directly affects the strength of spectral discrimination. When α is too small, the phase modulation effect becomes negligible, limiting the ability of the model to emphasize discriminative spectral components. Conversely, an excessively large α may introduce excessive phase distortion, which can adversely affect background reconstruction stability. As shown in Table IV, a moderate value of α achieves the best balance between spectral enhancement and reconstruction robustness.

The second one is loss balance coefficient λ. Fig. 10 shows the AUC curve when λ varies from 0.4 to 0.8. The model is relatively stable in [0.5, 0.7], and peaks at λ=0.6, which is hence adopted for all datasets.

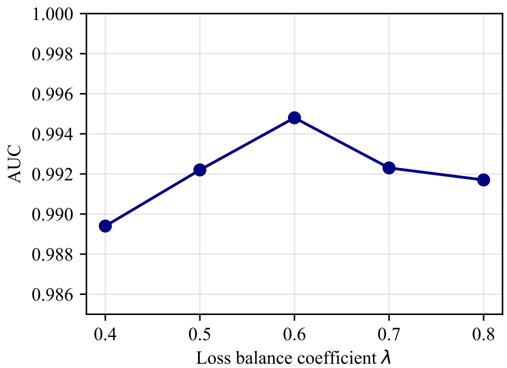

Fig. 10: The AUC curve when λ varies from 0.4 to 0.8 on Texas Coast.

We additionally evaluate the impact of different neighborhood sizes K used in graph construction. Extremely small K values limit the ability to capture contextual information, while overly large K values introduce background heterogeneity. Empirically, moderate neighborhood sizes yield the best performance, striking a balance between local consistency and global structure modeling.

In summary, the ablation study corroborates that every introduced module contributes positively to the final detection accuracy, and the full model achieves the best compromise between accuracy and computational overhead.

V. Conclusion

In this paper, we proposed a novel HAD framework that jointly addresses the challenges of spectral variability, spatial scale diversity, and insufficient feature discriminability in complex scenes. By integrating frequency-domain learning, adaptive spatial modeling, and structured representation regularization into a unified reconstruction-based paradigm, the proposed method significantly enhances both detection accuracy and robustness. Specifically, a LFPRB was introduced to perform adaptive spectral frequency modulation, enabling the network to selectively emphasize discriminative spectral components while preserving background stability. This frequency-aware representation effectively complements conventional spatial-domain processing. Furthermore, a MDCM was designed to generate input-dependent convolution kernels across different receptive fields, allowing the network to flexibly adapt to anomaly targets with diverse spatial scales and irregular structures. To further improve feature compactness and background consistency during training, a GCSFM was incorporated, which exploits multiscale neighborhood relations and contrastive constraints to enhance the separation between background and anomalous representations. Notably, this module is only activated during training and introduces no additional computational burden at inference time. Extensive experiments on multiple real hyperspectral datasets demonstrate that the proposed method consistently outperforms state-of-the-art approaches in terms of detection accuracy, background suppression, and robustness under complex background conditions.

Future work will focus on further improving the computational efficiency of large-scale hyperspectral inference, extending the framework to multi-temporal and cross-sensor anomaly detection scenarios, and exploring more lightweight frequency–spatial interaction mechanisms to enhance real-time applicability in operational remote sensing systems.

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